

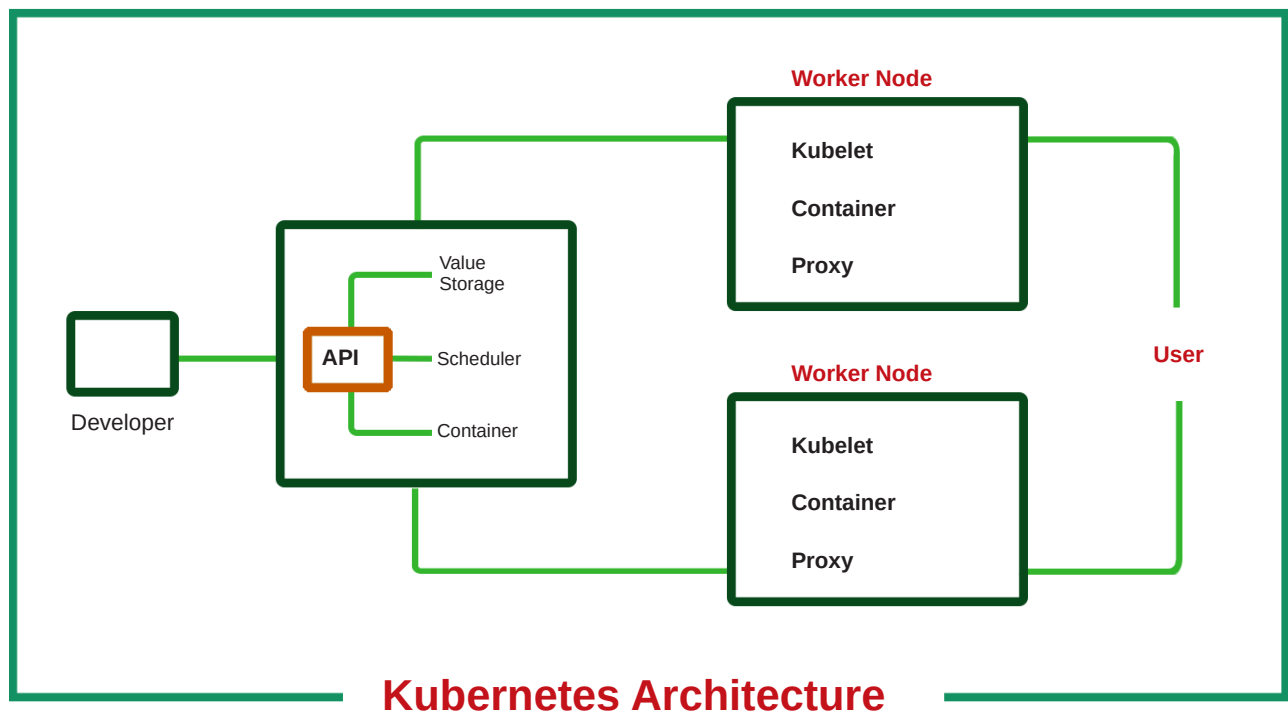


Figure 1: Kubernetes setup

## Kubernetes Architecture

service discovery and job scheduling, which occur with the deployment of microservices, it is often utilised as a host for such endeavours.

In addition to the original Kubernetes codebase, there are a number of cloud providers and distributions that provide their very own versions of the Kubernetes container orchestration system.

Here are a few Kubernetes attributes and qualities:

- Performs a number of hard tasks automatically, in addition to regulating server hosting and startup.
- Services like data storage, networking security, and management are controlled by containers.
- Maintains a constant vigil on the states of the nodes and containers under its care.
- Rollbacks are handled automatically for any modifications that are unsuccessful.
- Puts together an application mount for storage and adds a storage system to utilise with it.

### Why is Kubernetes considered imperative?

Kubernetes has seen increased demand as a result of the trend towards 'microservices', which are defined as applications that are smaller and more agile. Containers are the appropriate host for these decoupled microservices, since they are able to manage dependencies, operate independently, are not dependent on any particular platform, and may be destroyed when they are no longer required.

It is not unheard of for an application to be composed of hundreds, or even thousands, of microservices. This kind of program often has a great number of moving pieces. If we manage all of these varied components using custom-written programs or scripts, then increasing the size of these microservices while maintaining availability will be a challenge. Because of this, there is a need for a better technique of component management.

### A glance behind the scenes at how everything works

When a K8s cluster is in operation, one of the nodes is designated as the master, and it is this node that is responsible for making all the crucial decisions affecting the cluster. The other nodes, which are referred to as slaves or worker nodes, are responsible for carrying out the workloads for the applications in accordance with the master's directives.

### A straightforward Kubernetes design

Figure 1 presents an illustration of a fundamental K8s setup, which consists of a single master node and two worker nodes.

### Master node(s)

The master node acts as the cluster's primary coordinator. It is responsible for making decisions on the general status of the cluster as well as the duties that are delegated to each of the worker nodes. In order to make a master node operational and ready for usage, the following four operations need to be running on it.